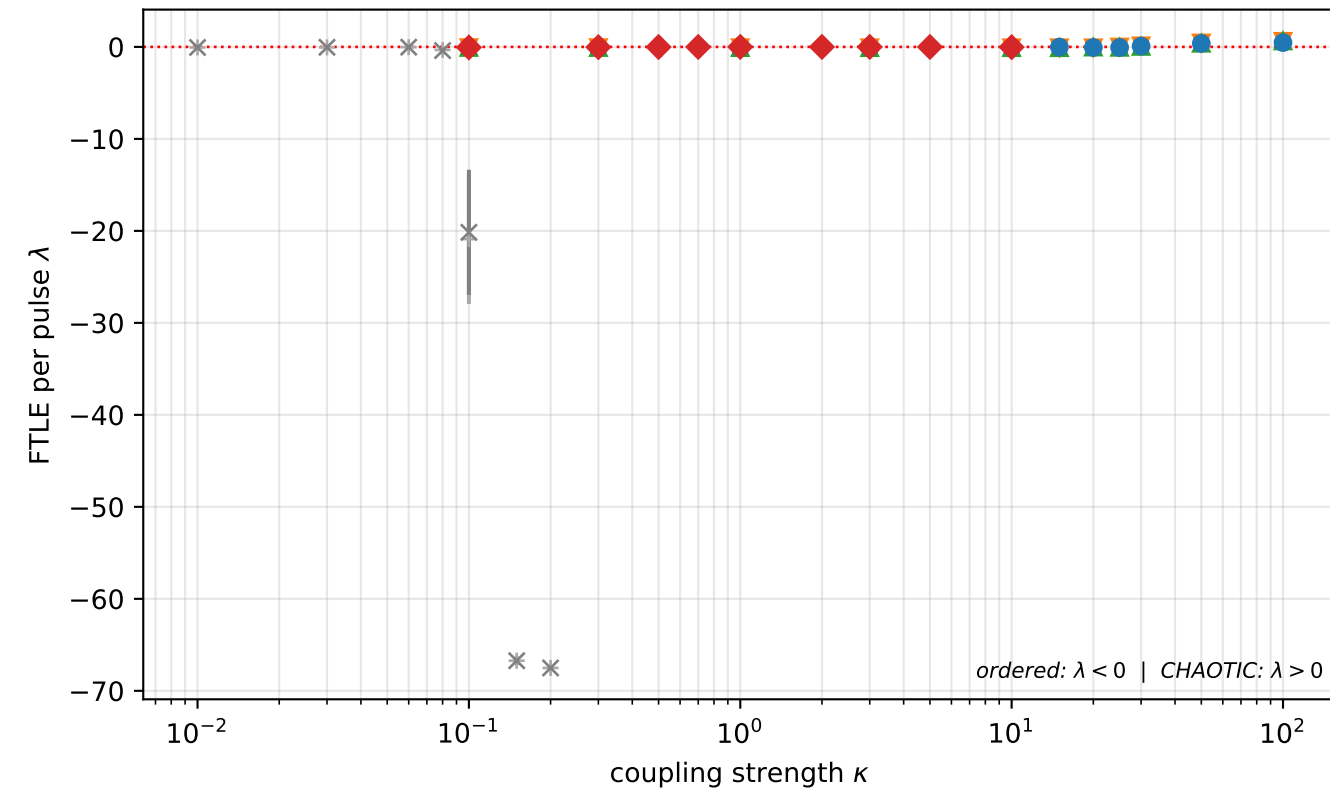


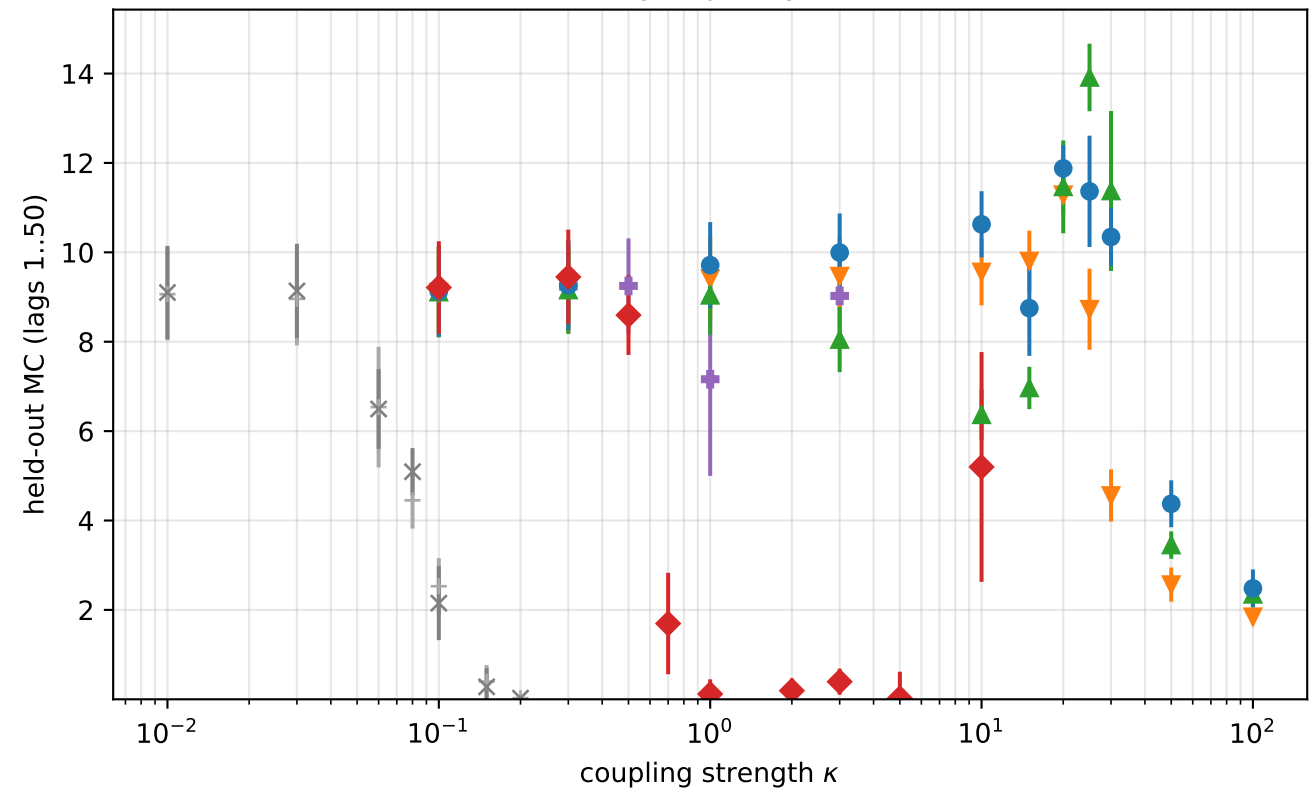
Substrate phase diagram v2: $N=256$, $CV=0.20$, $dt=0.2\mu s$, $20\mu s$, 10 seeds per config

■ parallel (uncoupled) ▴ lateral ring (mode 1) ◆ ring_unidir (mode 2) ✖ add_ring_bidir (mode 3, control)
 ● ring_bidir (mode 1) ▲ random_graph (mode 1) ◆ hub_star (mode 2) ✖ add_random_graph (mode 3, control)

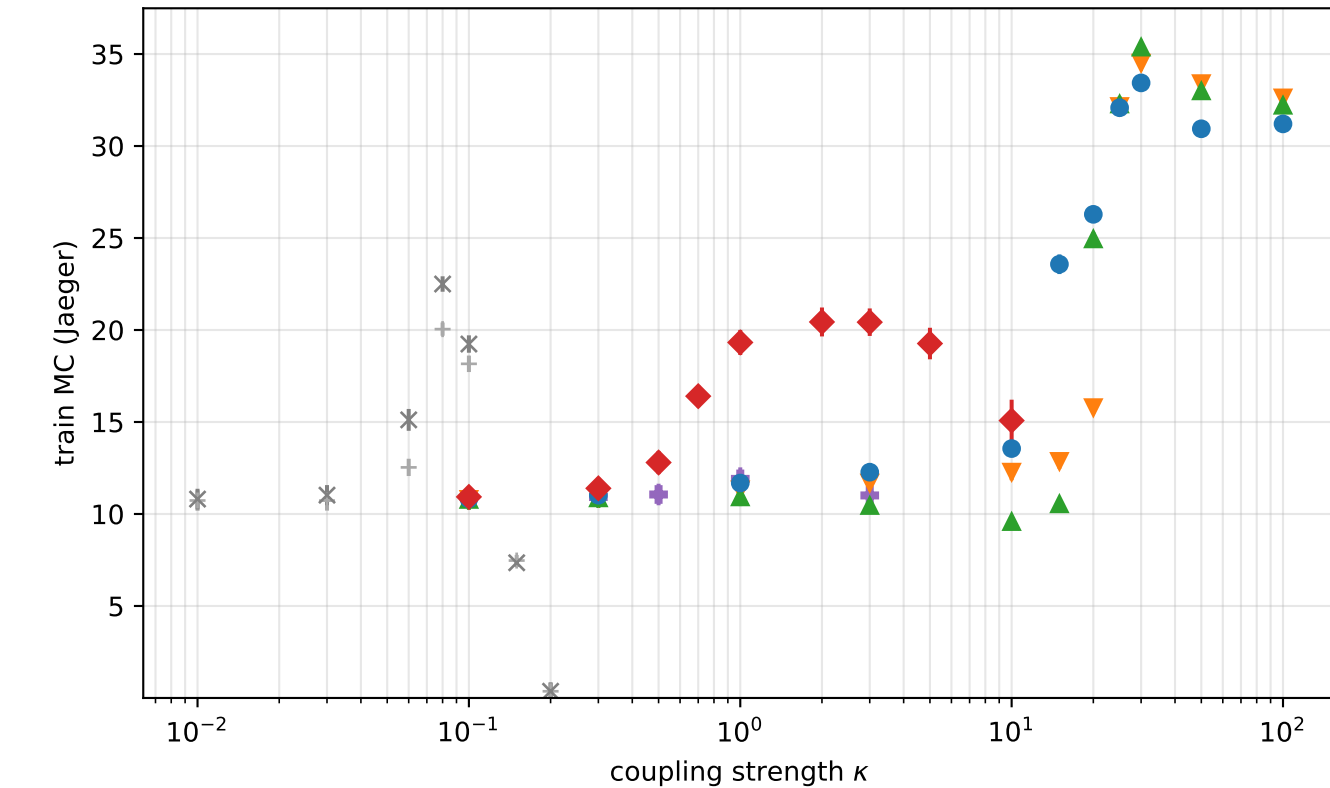
(a) Dynamics regime: edge of chaos at $\lambda \approx 0$



(b) Held-out memory capacity (honest estimate)



(c) Train-segment memory capacity (overfitting inflated in chaos)



(d) Saturation: clip fraction (self-limiting dynamics)

